ClaimsGuru™ Enterprise Platform

Backend Technology Stack & Azure Pre-Production Deployment Guide

Target Audience:	Marketing, Sales Enablement, Enterprise Buyers	**Environment:**	Microsoft Azure Pre-Production (Central India)
Live Domains:	https://www.wct-health.com | wct-health.com	**Resource Group:**	claimsGuru (All Services Provisioned)
Classification:	Enterprise Technical & Marketing Brief	**Verification Date:**	October 2026 (Live Audit & CLI Verified)

1. Executive Summary: What Powers ClaimsGuru?

ClaimsGuru is an intelligent, enterprise-grade health insurance adjudication and claim auditing platform. It eliminates manual claim verification by transforming multi-page, scanned hospital bills, diagnostic receipts, and discharge summaries into structured, verified, line-by-line financial tables and IRDAI-compliant audit reports.

Running natively on **Microsoft Azure**, the backend uses a modern **asynchronous microservices architecture** built with **Python FastAPI**, **Celery**, and **Redis**. AI extraction combines **Azure AI Document Intelligence** for computer vision table parsing with **Azure OpenAI GPT-4o** for clinical reasoning and coding, delivering turnaround times under 25 seconds with complete mathematical precision and zero data loss.

Pillar	Marketing Pitch & Enterprise Client Benefit
■ **High-Speed Async Engine**	Multi-page hospital bills (50+ pages) are processed in the background with zero UI lag via Celery queues and in-memory Redis message brokers.
■ **Dual-Engine AI Accuracy**	Pairs Azure Document Intelligence table coordinate extraction with Azure OpenAI GPT-4o parallel semantic parsing to capture complex hospital tariffs flawlessly.
■■ **Bank-Grade Security**	Secured via Microsoft Entra External ID (SSO/OAuth2), TLS 1.3 encryption, Azure Key Vault secret management, and 100% private Azure Blob Storage.
■ **Elastic Cloud Scalability**	Containerized microservices running on Azure Container Apps autoscale dynamically from 1 to 10 instances based on concurrent claim load.

2. Backend Technologies Used (Explained in Simple Terms)

Technology Layer	Selected Technology	Simple Explanation & Why We Use It
Core API Gateway	**FastAPI & Python 3.11** ASGI Engine: Uvicorn	Ultra-fast, asynchronous REST API. Handles user uploads and claim lifecycle with sub-second response times without blocking server resources.
Background Workers	**Celery 5.4** Broker: Redis 7	Asynchronous task engine. Offloads heavy AI and OCR computing from the website so the user's browser never hangs or buffers.
Real-Time Cache	**Redis 7 Container** In-Cluster TCP 6379	Ultra-fast in-memory cache. Powers real-time live progress bars (0% to 100%) and instant session retrieval for auditors.
Generative AI / LLM	**Azure OpenAI GPT-4o** Global (440K TPM)	Enterprise-hosted Multimodal AI. Parses messy pharmacy items, extracts clinical diagnoses, and suggests accurate ICD-10 and CPT codes.
Computer Vision OCR	**Azure Document Intelligence** (Form Recognizer S0)	Deep layout analysis. Detects table boundaries, columns, headers, and text from blurry scanned hospital receipts and discharge summaries.
Relational Database	**Azure SQL Database** (Standard S1 - 20 DTUs)	Bank-grade relational database. Safely stores claims, hospital rosters, itemized bills, policy rules, and immutable audit logs.
Secure File Lake	**Azure Blob Storage** Hot Tier (claimgpt)	Encrypted object storage. Stores raw medical files and generated PDFs with zero public internet access (access only via secure token).
Enterprise Identity	**Microsoft Entra ID** (CIAM / OAuth 2.0)	Enterprise Single Sign-On (SSO) and authentication. Ensures only authorized employees and auditors can access patient health records.
Automated PDF Engine	**WeasyPrint & Jinja2**	High-fidelity PDF generation. Generates official IRDAI Standard Claim Forms (Part A & B) and executive TPA reimbursement summaries.

3. Live Pre-Production Cloud Services & Configurations

ClaimsGuru pre-production is fully deployed in Microsoft Azure within the **claimsGuru** resource group (Central India). The environment mirrors enterprise production architecture with automated autoscaling, private data isolation, and redundant services.

Service / Resource Name	Azure Type	Sizing & Specs	Autoscaling / Configuration	Role in Platform
cg-preprod-cin-frontend	Azure Container App	0.5 vCPU 1.0 GiB RAM	Autoscales: **1 – 5 replicas** Custom Domain: wct-health.com	Next.js 14 web application. Auditor Cockpit, customer dashboard, and landing page.
cg-preprod-cin-ingress	Azure Container App	1.0 vCPU 2.0 GiB RAM	Autoscales: **2 – 10 replicas** Port 8000 (External HTTP)	FastAPI API Gateway. Uploads, Entra JWT auth, claim CRUD, and status polling.
cg-preprod-cin-worker-default	Azure Container App	1.0 vCPU 2.0 GiB RAM	Autoscales: **2 – 8 replicas** Celery default queue	Background worker. Clinical coding, IRDAI rule checks, anomaly checks, and PDF creation.
cg-preprod-cin-worker-ocr	Azure Container App	1.0 vCPU 2.0 GiB RAM	Autoscales: **1 – 6 replicas** Celery ocr_queue	Dedicated vision worker. PDF rasterization and Azure Document Intelligence extraction.
cg-preprod-cin-redis	Azure Container App	0.5 vCPU 1.0 GiB RAM	Static: **1 replica** Internal Port 6379	In-memory task broker for Celery queues and live claim progress bar synchronization.
cg-preprod-sql-vvekhbufuteq2 ***(DB: `claimsguru`)***	Azure SQL Database	Standard Tier (20 DTUs)	Max 250 GB storage Max 60 concurrent workers	Transactional relational persistence for claims, tariffs, line items, and audit trail.
cgpreprodstvvekhbufuteq2 ***(Container: `claimgpt`)***	Azure Blob Storage	StorageV2 Hot (Standard LRS)	Public Access: **DENIED** Enforce HTTPS (TLS 1.2+)	Secure document lake storing uploaded claim PDFs, bills, and generated audit reports.
cg-preprod-openai	Azure OpenAI	GPT-4o (2024-11-20) Standard S0	**440K Tokens/Min** Global Standard routing	Generative AI engine for semantic table parsing and clinical reasoning.
cg-preprod-docintel-...	Azure Doc Intelligence	Standard S0 (Form Recognizer)	Prebuilt Read & Layout Direct REST API integration	Computer vision OCR extracting bounding boxes, tables, and handwriting.
cg-preprod-sb-vvekhbufuteq2	Azure Service Bus	Standard Tier	Queues: default, ocr_queue	Enterprise cloud message bus guaranteeing zero message loss during traffic spikes.
cg-preprod-kv-vvekhbufut	Azure Key Vault	Standard Tier	Soft-delete enabled RBAC Secret Access	Hardware-backed vault storing all database passwords, API keys, and connection strings.
cg-preprod-comm **cg-preprod-email**	Azure Communication	Enterprise Global Pay-As-You-Go	Verified Domain: notifications.wct-health.com	Automated transactional email dispatch sending audit notifications to patients and auditors.
cg-preprod-cin-law	Azure Log Analytics	PerGB2018 Tier	30-Day Retention Centralized Workspace	24/7 telemetry, container metrics, and regulatory audit log tracking.
acrclaimsgurupreprod	Azure Container Registry	Basic Tier (10 GiB included)	Private Repository Active tag: preprod-rev104	Private Docker registry storing and deploying versioned container microservices.

4. The End-to-End Claim Journey: 5 Simple Steps

From the moment an employee or hospital uploads documents to the final adjudication report delivery, every step runs automatically and seamlessly:

Step 1: Secure Ingestion (0–2s)	Claimant signs in securely via Microsoft Entra External ID and uploads medical bills. Ingress streams files directly to encrypted Azure Blob Storage.
Step 2: Table & Layout OCR (2–8s)	OCR background worker extracts tables, line items, hospital letterhead, and handwriting using Azure AI Document Intelligence computer vision.
Step 3: Clinical AI Coding (8–18s)	Azure OpenAI GPT-4o parses complex medical terms, maps diagnosis to standard ICD-10 and CPT codes, and checks consistency with doctors' notes.
Step 4: IRDAI Adjudication (18–22s)	Automated rules engine scores the claim against Indian IRDAI guidelines, identifies non-payable items, and runs fraud and duplicate checks.
Step 5: Reports & Notifications (22–25s)	System generates official IRDAI Standard Form and TPA Executive Adjudication PDFs, updating the Auditor Cockpit and emailing stakeholders.

5. Enterprise Security, Privacy & Reliability Highlights

Key security pillars ready for enterprise procurement, CISO review, and sales presentations:

Security Dimension	Enterprise Guarantee & Implementation
■■ Data Privacy & Isolation	Zero public access to patient files. All medical records in Azure Blob Storage are private and accessed strictly via authenticated, temporary tokens.
■ End-to-End Encryption	TLS 1.3 encryption in transit for all web and API traffic. Microsoft AES-256 hardware-level encryption at rest across databases and storage.
■ Hardware Key Vault	Zero passwords, connection strings, or API tokens stored in code. All credentials retrieved securely from Azure Key Vault with soft-delete enabled.
■ High Availability & Resilience	Microservices autoscale up to 10 instances during traffic surges. Asynchronous message queues ensure zero data loss even during peak billing cycles.
■ Regulatory Audit Trail	Every AI extraction, tariff adjustment, and claim status change is permanently timestamped and logged for regulatory transparency and IRDAI compliance.

Verification Note: All technical specifications, Azure resource names, container sizing, and domain configurations in this document have been directly verified against the active claimsGuru pre-production environment in Microsoft Azure.